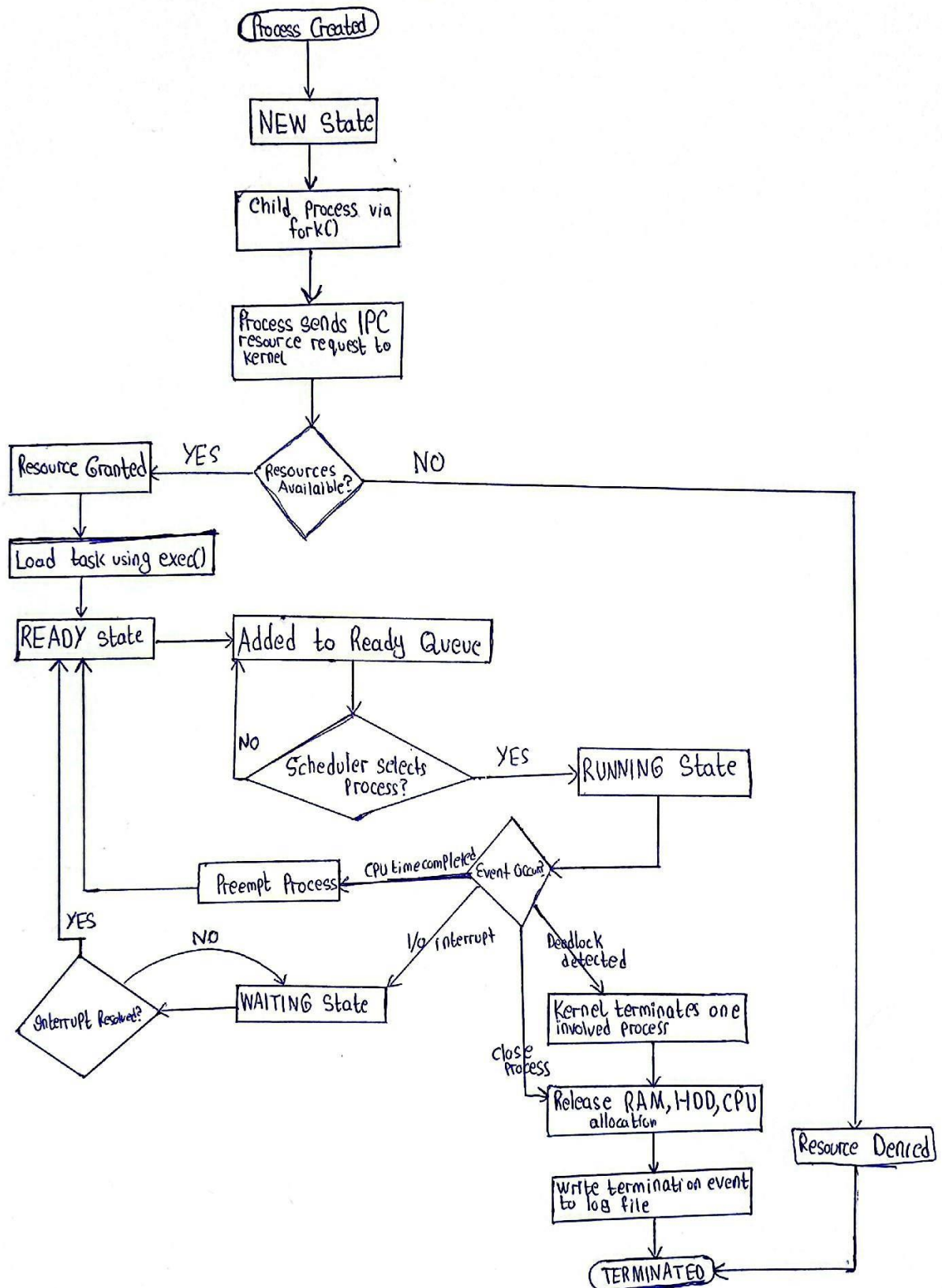
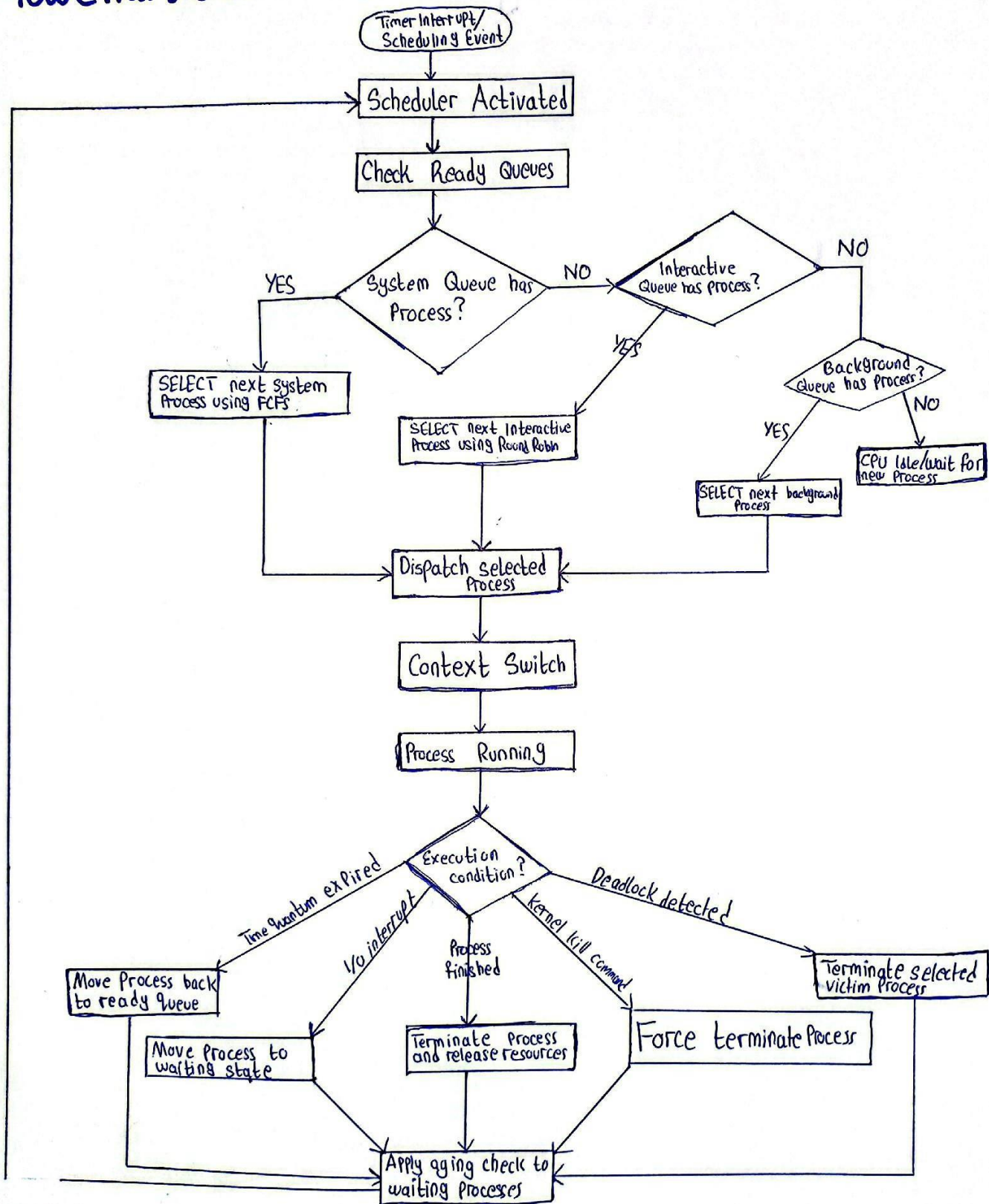


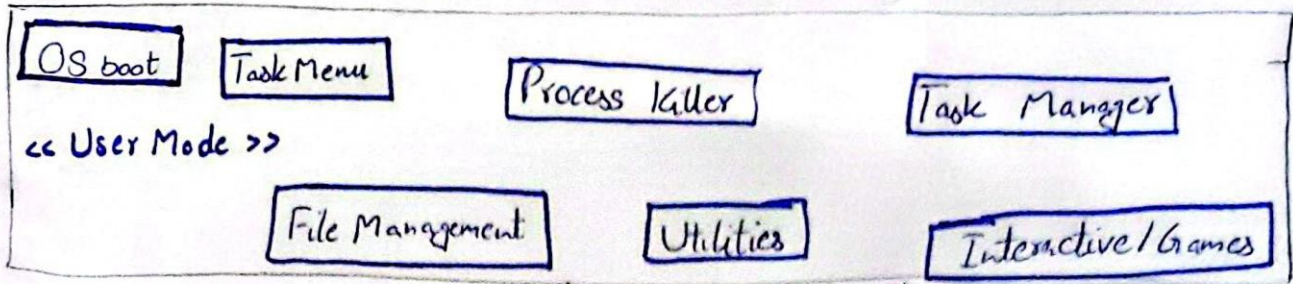
FlowChart (Process Lifecycle)



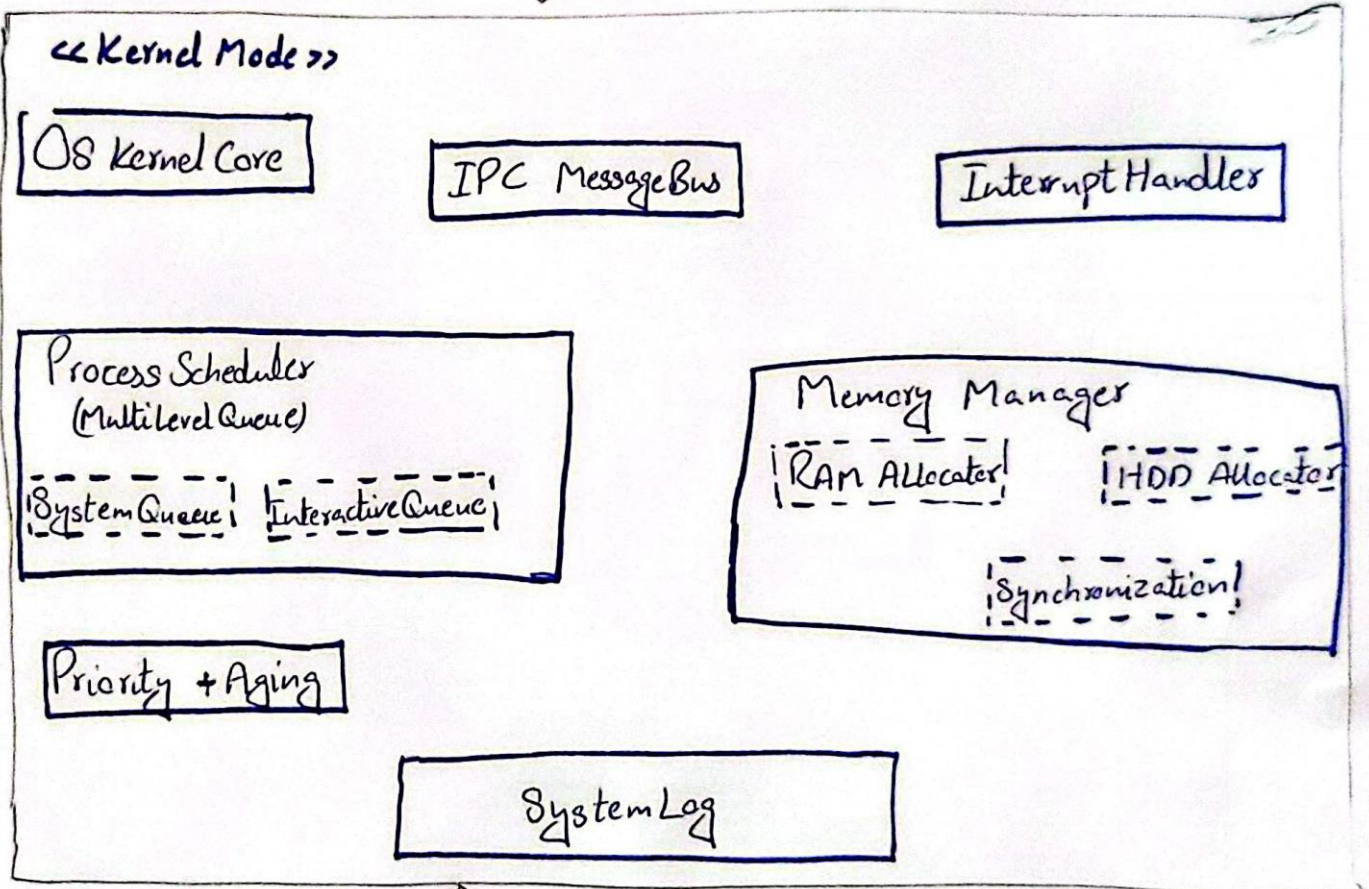
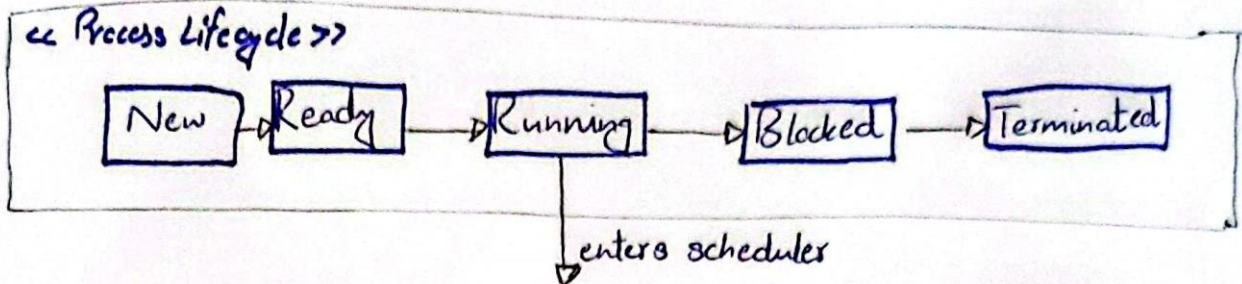
FlowChart (Scheduling)



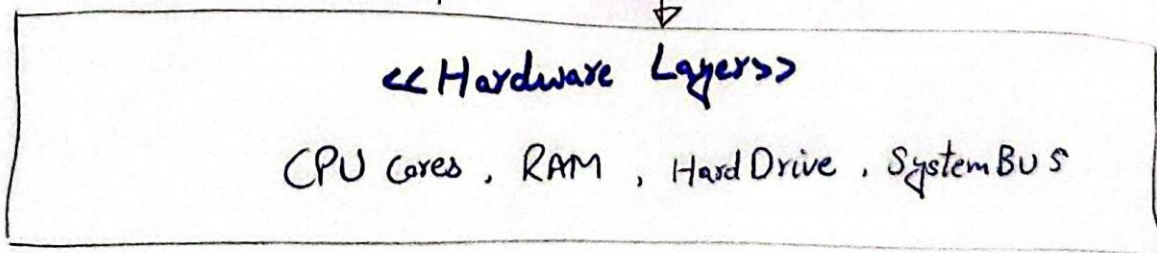
System Architecture Diagram



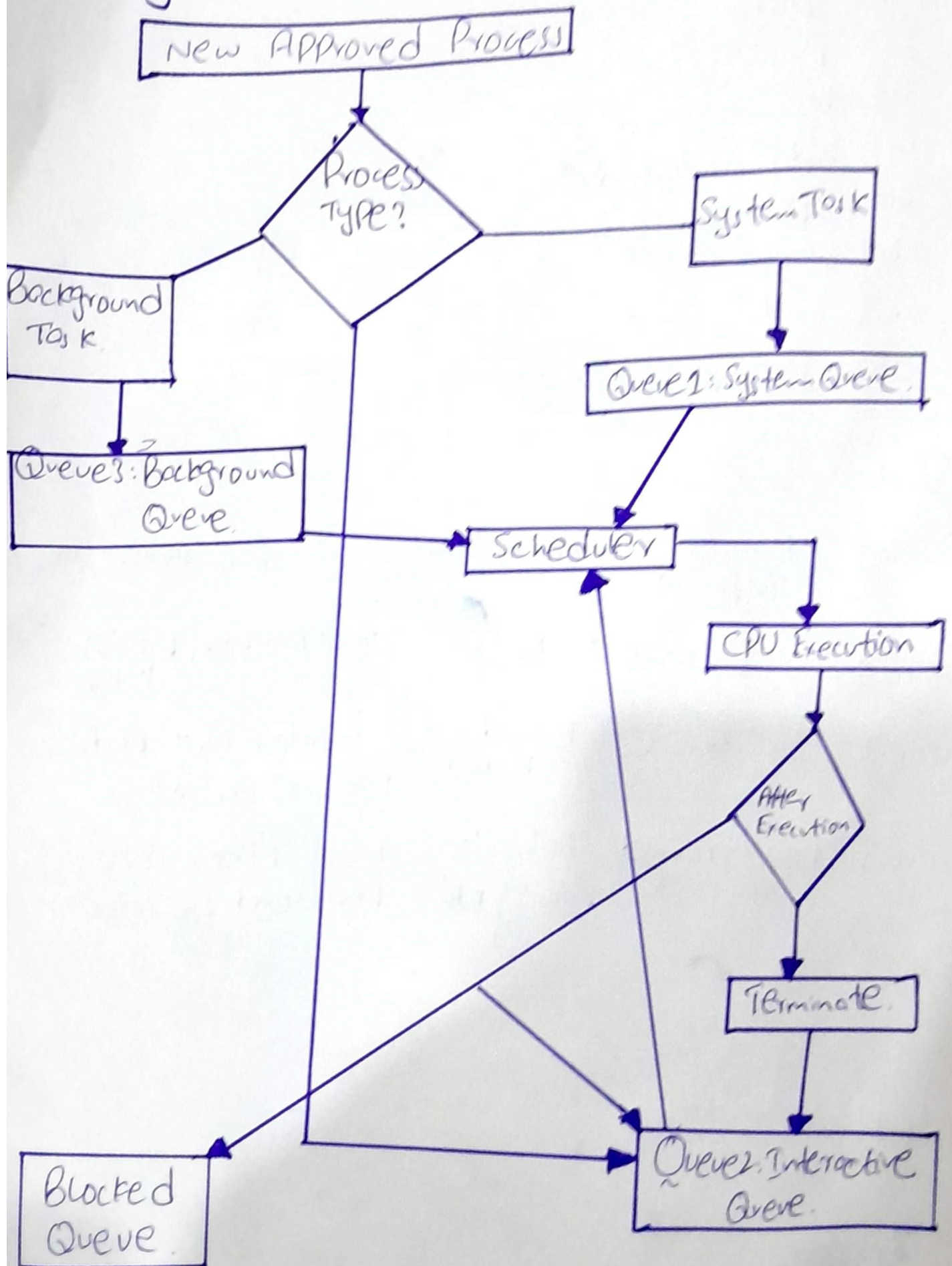
Launch task / interrupt



sys call / Hardware Access



Ready Queue Design:



struct Process {

int pid;

int priority;

int ram-required;

int hdd-required;

int cpu-required;

int state;

int waiting-time;

char process-type[20];

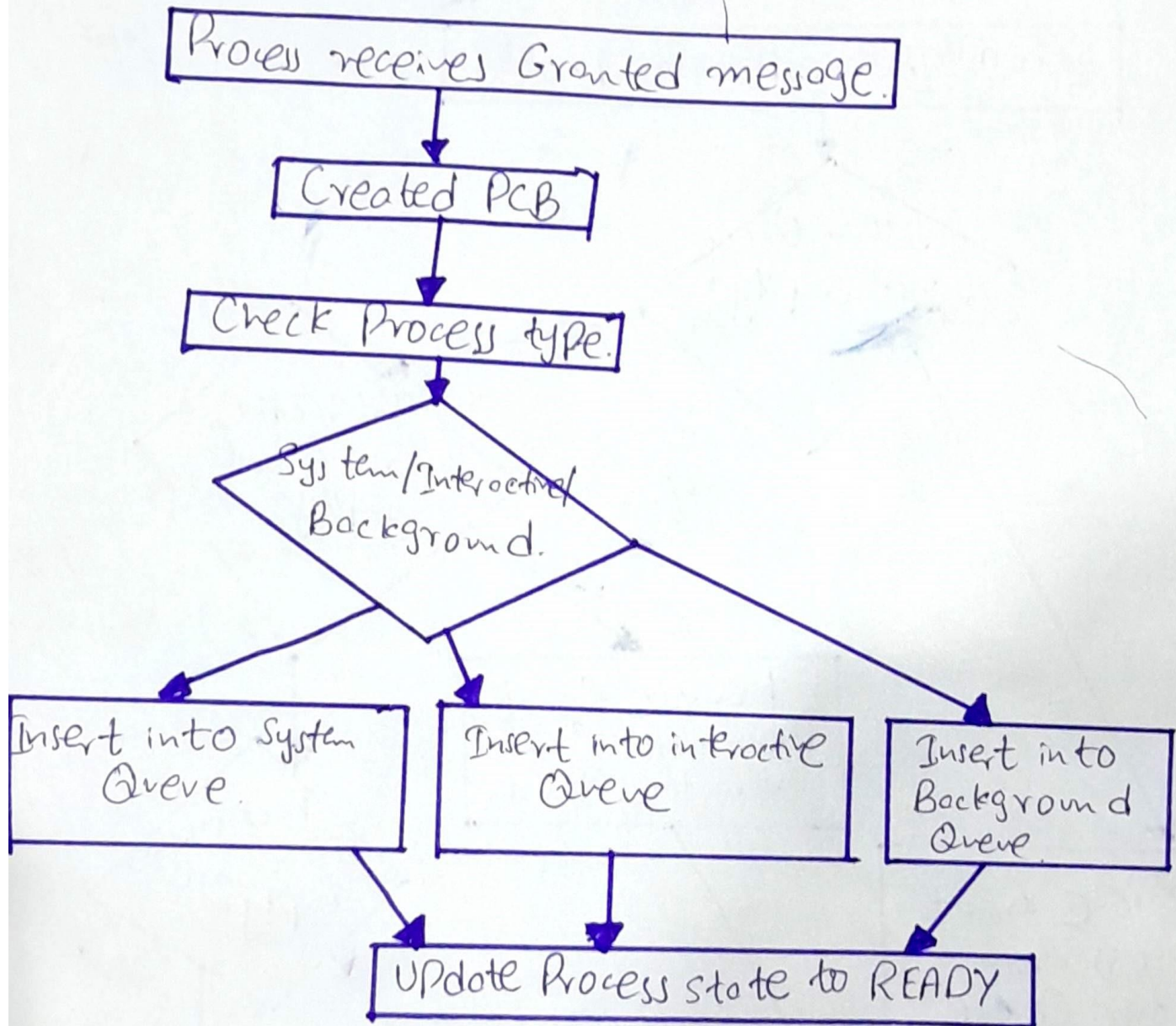
};

Queue levels:

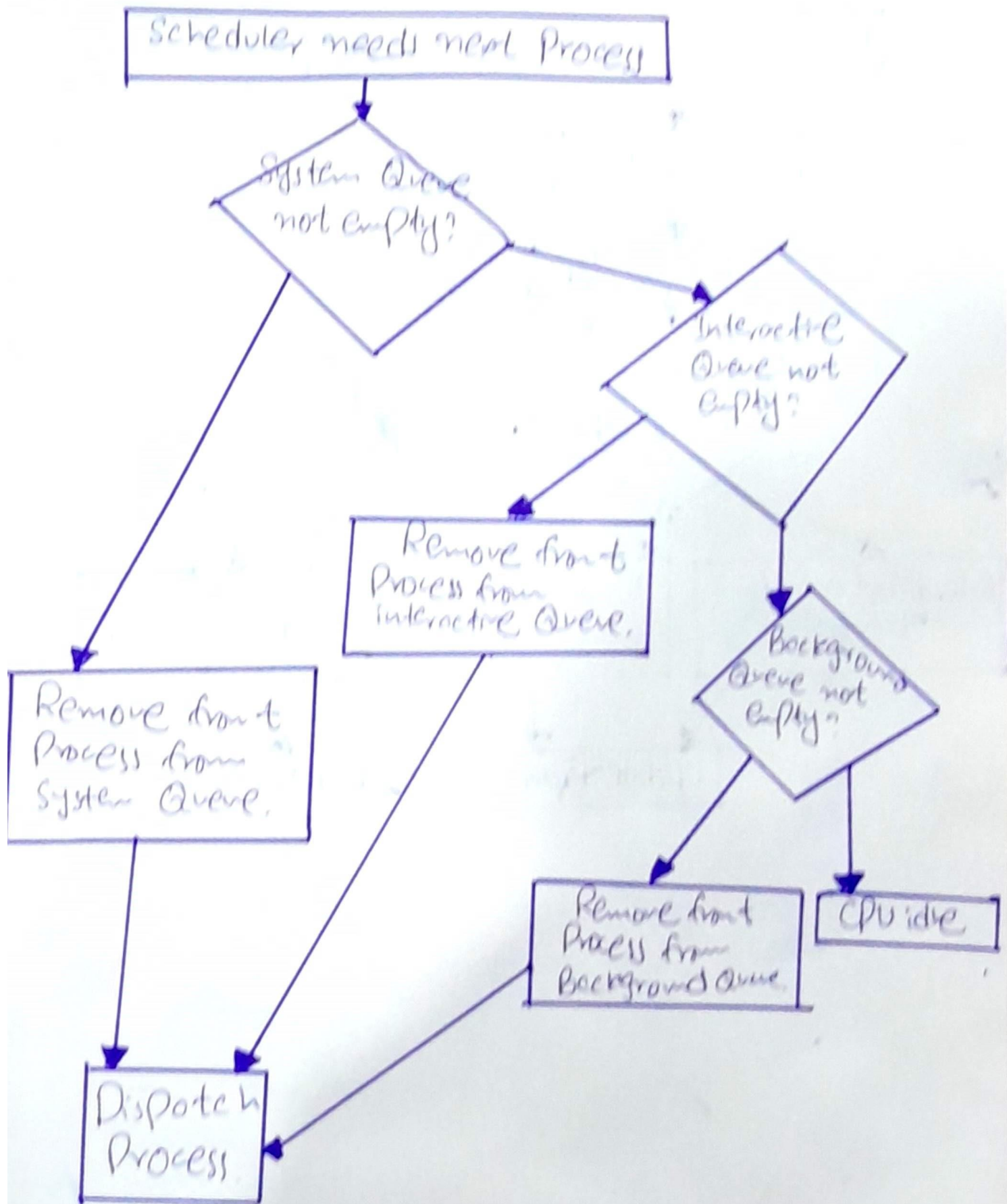
Queue	Process Type	Algorithm	Example Tasks.
Queue 1	System-level	FCFS	Task Manager, Process Killer.
Queue 2	Interactive	Round Robin	Calculator, Notepad, File viewer, Games.
Queue 3	Background	FCFS / Low Priority RR	Music Player, File Download Simulator.

Add Process to Queue:

(2)

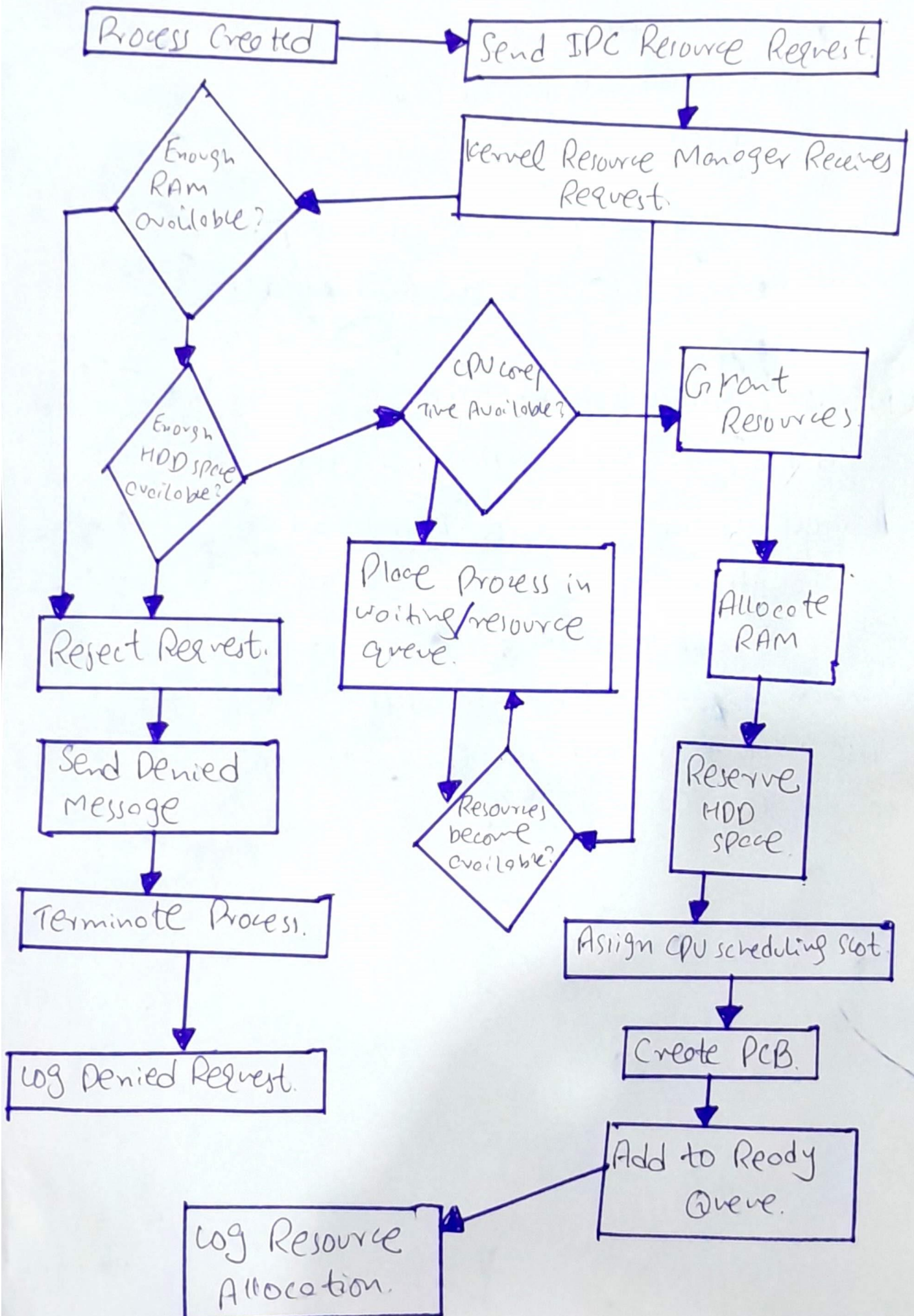


Remove Process from Queue



Resource Allocation Strategy:

3



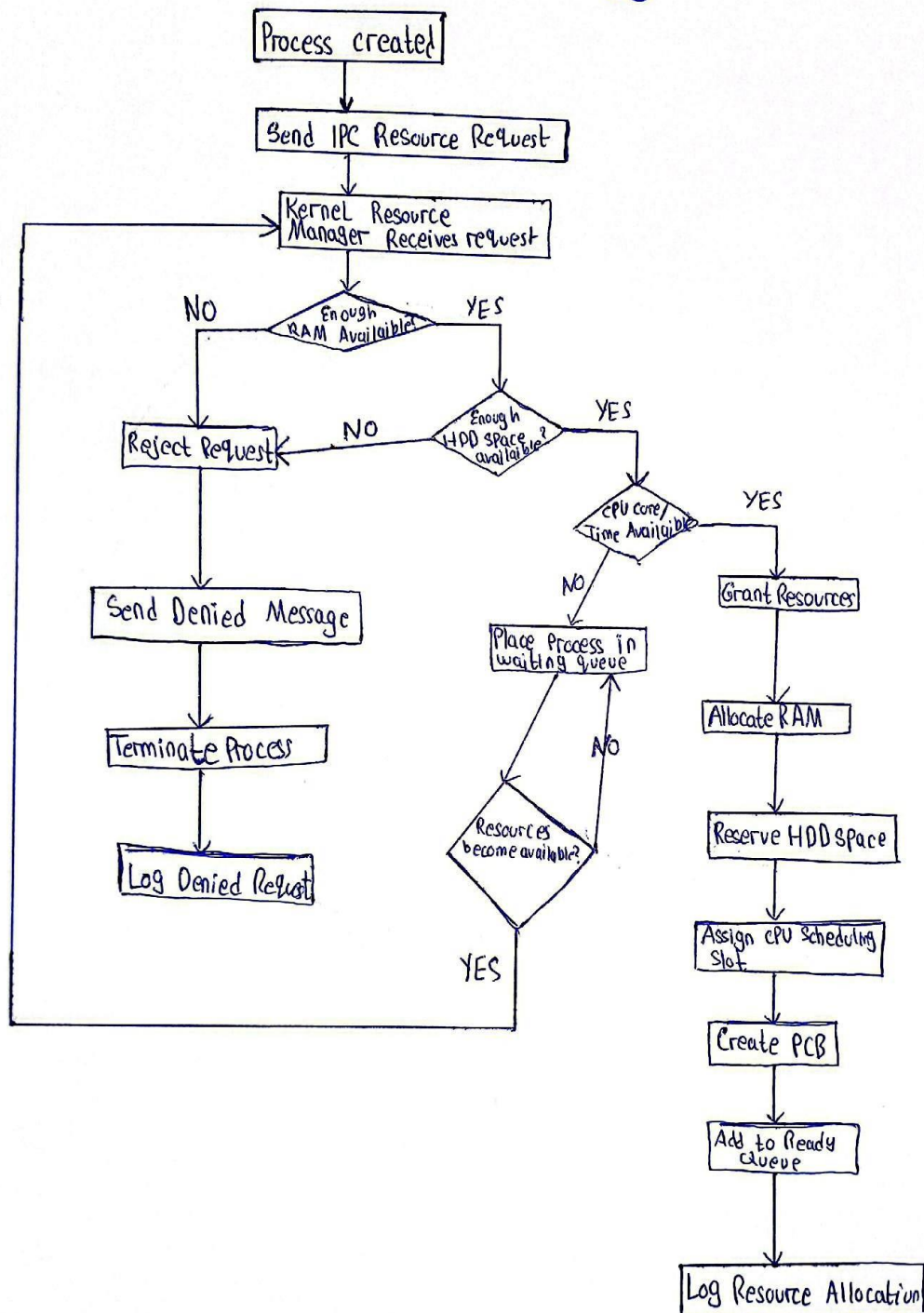
CPU Allocation Policy:

- 1- System tasks get highest queue Preference
- 2- Interactive tasks use Round Robin for fairness.
- 3- Background tasks run only when higher queues are empty or waiting.
- 4- Aging increases Priority of long waiting Processes.
- 5- Deadlock victim Processes are terminated to free resources.

Memory Allocation Policy:

- 1- Each process requests RAM before execution.
- 2- Kernel checks available RAM.
- 3- If enough RAM exists, memory is allocated.
- 4- If not enough RAM exists, request is denied.
- 5- When a process terminates, RAM is released.
- 6- All allocation and release events are written to the log file.

Resource Allocation Strategy



CPU Allocation Policy

1. System tasks get highest queue preference
2. Interactive tasks use Round Robin for fairness
3. Background tasks run only when higher queues are empty or waiting
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